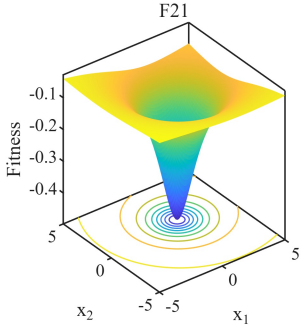
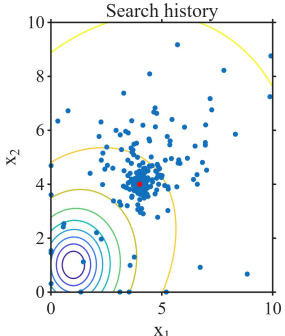
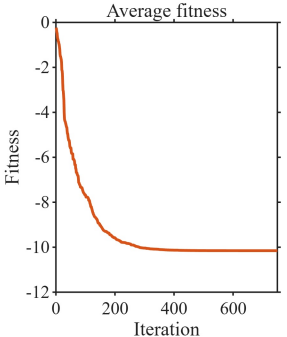
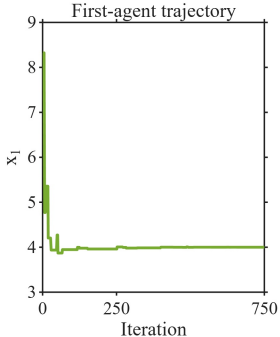
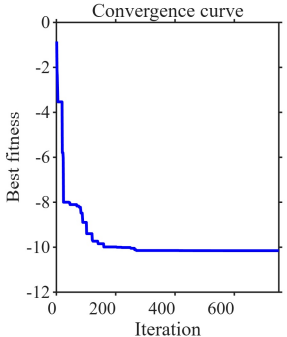

F21
Fitness
-0.1
-0.2
-0.3
-0.4
5
0
-5
x2
-5
0
5
x1


Search history
x2
10
8
6
4
2
0
0
5
10
x1


Average fitness
Fitness
0
-2
-4
-6
-8
-10
-12
0
200
400
600
Iteration


First-agent trajectory
x1
9
8
7
6
5
4
3
0
250
750
Iteration


Convergence curve
Best fitness
0
-2
-4
-6
-8
-10
-12
0
200
400
600
Iteration